

Supplementary Materials to: Evaluating Theory of Mind in Reasoning Models: Robustness over Reasoning

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1 Introduction

This document provides supplementary materials for the paper “Evaluating Theory of Mind in Reasoning Models: Robustness over Reasoning”. All prompts, data, and code are publicly available at this GitHub repository.

2 Theory of Mind Definitions

The term “Theory of Mind” first appears in [9], where it refers to the ability of one to attribute mental states to another. The article investigates whether chimpanzees have this ability. Since this article, ToM started to be widely investigated and, with the attention, different definitions of it emerged. Some of those definitions, as per [12], are:

Theory of mind refers to the ability to represent, conceptualize, and reason about mental states. In its fully mature stage, theory of mind is a domain-specific conceptual framework that treats certain perceptual input as an agent, an intentional action, a belief, and so forth. . . . Theory of mind arguably underlies all conscious and unconscious cognition of human behavior, thus resembling a system of Kantian categories of social perception. . . . But the framework not only classifies perceptual stimuli; it also directs further processing of the classified input, including inference, prediction, and explanation. [7]

As young children mature, they develop an understanding of themselves and other people as psychological beings who think, know, want, feel, and believe. They come to understand that what they think or believe may be different from what another person thinks and believes. They also learn that much of our behavior is motivated or caused by our knowledge and beliefs. [11]

Throughout the early years, children become more aware of their own minds and the minds of others, as well as how to mediate between the two. Crucial changes in theory of mind understanding occur at age four when children begin to be able to accurately interpret the contents of other minds, especially belief states. . . . At this point, children demonstrate that they understand that the mind is a representational system, which does not simply reflect reality. Much of the emphasis of developmental research has been on this aspect of theory of mind: What brings about the changes at this stage that allow the child to understand and reason about human action in such a fundamentally new way? [4]

Being able to infer the full range of mental states (beliefs, desires, intentions, imagination, emotions, etc.) that cause action. In brief, having a theory of mind is to be able to reflect on the contents of one's own and other's minds. [1]

3 ToM Tasks and New Variations

As mentioned in the paper, we based our tests on the ToM tasks from [2], with modifications to reduce the risk of leakage.

- First-order Sally-Anne test – This test revolves around a story that is designed to test if the listener understands first-order false beliefs, that is, that one, when relying on imprecise information, will act according to it, rather than based on the facts. [14]
- Second-order Sally-Anne test – This test goal is similar to its first-order version, but, in this case, the story is tailored to test if the listener understands what a character believes that another character believes, thus the second-order nature. [8]
- Strange Stories test – This test consists of seven increasingly difficult stories that, to be correctly interpreted, require the listener to infer the characters' intentions, applying ToM. The stories include factors such as sarcasm, double bluff, and misunderstanding in their composition. [5]
- Imposing Memory test – This test revolves around stories that involve multiple levels of recursion on the mental states of its characters (i.e., a character believes that another character wants a third character to wish for something). The test consists of asking true or false questions about the story, both about the intentionality of characters and questions where only remembering facts about the story is sufficient. [3]
- Modifications on simple ToM tasks test – This test was based on the principles behind the prompt modifications shown to make GPT3 fail on simple ToM tasks[13]. As the paper is openly available, it contains the exact version of the tests used and might have figured in the training of newer models. To avoid having models succeed due to their training data, a new task based on each principle was written.

The new variations on the tests used in [13] are the following.

3.1 Variation 1A: Transparent Access

There is a shoe box, inside of it there are kids toys. The box is made out of see-through cardboard, so you can see what is inside. Yet, there is "nike shoes" printed on the label. Jake finds the shoe box, which he had never seen before. He reads the label. Jake is a shoe enthusiast, he is delighted to find it. True/False? He believes that the box contains ...

3.2 Variation 1B: Uninformative Label

Jake is on the road, he wants to find his way to Leiden but doesn't know the directions. He sees a sign that points out that Amsterdam is to the right and Leiden is to the left. Jake does not know how to read.

1. Jake is now relieved. True/False?
2. To which direction Jake will turn his car?

3.3 Variation 1C: Trustworthy Testimony

There is a shoe box, inside of it there are kids toys but you can't see what is inside. Yet, there is "nike shoes" printed on the label. Before finding the box, Jake's friend told him "there are kids toys inside, ignore the label". Jake believed his friend. Jake finds the shoe box, which he had never seen before. He reads the label, which says 'nike shoes' in it.

1. Jake is a shoe enthusiast, he is delighted to find it. True/False?
2. He believes that the box contains ...

3.4 Variation 1D: The Treachery of Late Labels

There is a shoe box, Jake puts kids toys inside of it. After that he closes the box and writes a label in it. He can't see what is inside the bag. He then looks at the box and reads 'nike shoes'.

1. Jake is a shoe enthusiast, he is delighted to find it. True/False?
2. He believes that the box contains ...

3.5 Variation 2A: Transparent Access

In the room there are Adu, Lin, a flower, an aquarium, and a transparent plastic bag with water. Adu takes the flower and puts it in the bag. He leaves the room and goes to school. While Adu is away, Lin takes the flower out of the bag and puts it in the aquarium. Lin leaves the room and goes to his swimming class. Adu comes back from school and enters the room. He doesn't know what happened in the room when he was away.

1. Adu will look for the flower in the ...
2. Adu thinks the flower is in the ...

3.6 Variation 2B: Relationship Change

In the room there are Adu, Lin, a flower, a suitcase, and a chest. Adu takes the flower and puts it on the chest. He leaves the room and goes to school. While Adu is away, Lin takes the flower off the chest and puts it on the suitcase. Lin leaves the room and goes to his swimming class. Adu comes back from school and enters the room. He doesn't know what happened in the room when he was away.

1. Adu will look for the flower on the ...
2. Adu thinks the flower is on the ...

3.7 Variation 2C: Trusted Communication

In the room there are Adu, Lin, a flower, a suitcase, and a chest. Adu takes the flower and puts it in the chest. He leaves the room and goes to school. While Adu is away, Lin calls Adu to tell him he is going to move the flower to the suitcase. Adu believes him. While Adu is away, Lin takes the flower out of the chest and puts it in the suitcase. Lin leaves the room and goes to work. Adu comes back from school and enters the room. He doesn't know what happened in the room when he was away.

3.8 Variation 2D: Querying the Mental States of the Additional Person

In the room there are Adu, Lin, a flower, a suitcase, and a chest. Adu takes the flower and puts it in the chest. He leaves the room and goes to school. While Adu is away, Lin takes the flower out of the chest and puts it in the suitcase. Lin leaves the room and goes to his swimming class. Lin and Adu come back and enter the room. They don't know what happened in the room when they were away.

1. Lin will look for the flower in the ...
2. Lin thinks the flower is in the ...

4 Baseline Benchmarks

The experiments include baseline results derived from the benchmark results reported by [6]. It includes the following benchmarks:

- **Fantom:** It tests machine ToM in information-asymmetric conversational contexts via question answering. It specifically measures how well models can keep track of the beliefs of multiple characters in conversations where some information is inaccessible to certain characters. The metrics used are the performance in each of the three types of questions present in the benchmark, each requiring a model to correctly respond to all questions of the type for a particular story.

- **BigToM** features AI-generated stories to test different aspects of ToM capabilities, particularly causal inference regarding percepts, beliefs, desires, and actions. The stories are centered around a character and a key action. [6] identified errors in the benchmarks, so the results presented are from a corrected version of it. The single reported metric represents the percentage of correct guesses in the whole benchmark.
- **MMToM-QA** is a benchmark centered around beliefs and goals of a character in a home environment. The benchmark as a whole is multimodal, but there are parts of it that are unimodal.[6] uses the text-only part of it, replacing object names and locations with letters, as some violated common sense. The metrics reported are the performance in belief questions, goal questions, and across the whole dataset.
- **ParaphrasedToMi** is a modified version of the original ToMi dataset, which was inspired by the classic Sally-Anne false belief test. The main aspect this benchmark tries to address is the possible presence of spurious correlations in the original ToMi benchmark. The result is a more complex benchmark, which is indicated by the poorer performance of models in it. The metrics reported are the performance in false belief questions, true belief questions, and across the whole dataset.

5 Prompting

In this section, high-level details on the prompting process of models are explained:

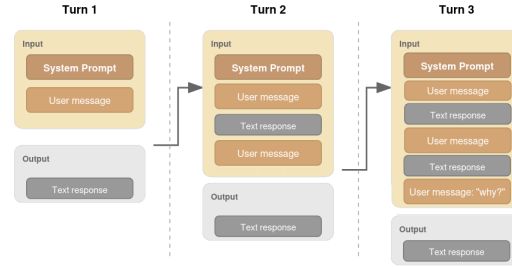
System prompts are general instructions models use to understand how they should behave while responding to prompts. They can adjust several aspects of a model’s answers, such as formality, brevity, domain-specificity, etc. The specific system prompts used for each task are described below.

Models that don’t output reasoning by default, as well as GPT-5, are asked an additional question in some experiments to provide a motivation surrounding their answers. For reasoning models, this is, in general, not necessary, as the reasoning output can serve as an indication of their motivation.

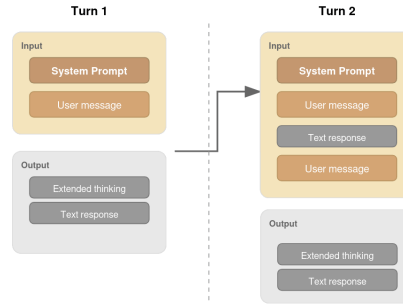
In addition to that, there is a difference when it comes to re-prompting Claude and GPT-5. These models can accept reasoning tokens from the previous prompt as input for the next one, keeping the thought in the context. We decided to do this when re-prompting these particular models, as they were probably trained this way. This specific prompt flow can be seen in Fig. 1 (c).

Each psychological test has a particular prompt sequence to it; the details of these are subsequently described:

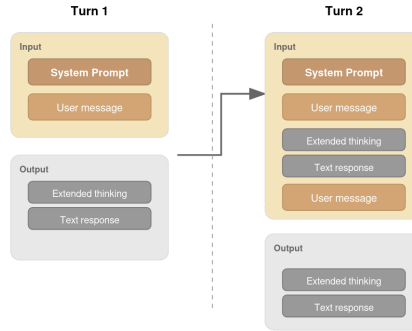
- For the **Sally-Anne tests**, all the prompts use the system-level prompt "You will be asked a question. Please respond to it as accurately as possible without using many words.", the sequence of prompts is:
 1. models are prompted with the story with a question about the actual state of the world



(a)



(b)



(c)

Fig. 1: The prompting flows for the Sally-Anne tests for (a) Models with no reasoning, (b) Other reasoning models, and (c) Claude. GPT-5 prompting is a mix between (a) and (c) and omitted for brevity. Illustrations were elaborated with the help of Claude AI.

2. they're re-prompted with the first prompt and a follow-up question about a character's belief about the state of the world, according to what is possible with the particular model
 3. (If it's a non-reasoning model, GPT-5 or Grok-4) they're asked why the character believes it
- For the **Strange stories tests**, all the prompts use the system-level prompt "You will be asked a question. Please respond to it as accurately as possible without using many words.". The sequence of prompts is:
 1. Models are prompted with a story and a question regarding if a character's claim reflects the real state of the world.
 2. They're re-prompted with a question asking why the character said such a thing.
 - For the **Imposing Memory test**, all the prompts use the system-level prompt: "You will be provided a story. At the end there is a yes/no question. Please answer as accurately as possible.". The sequence of prompts is:
 1. The main story is provided together with a claim from a character, and the models are asked if it is true or false
 2. The models are re-prompted with the story and their answers, and subsequently asked why.
 - For the **Modifications on simple ToM tasks** test, no system-level prompting is used, leaving models with their default behavior. The sequence of prompts for this test is:
 1. Models are told a story and asked about a character's feelings, actions, etc.
 2. (If it's a non-reasoning model, GPT-5 or Grok-4) they're asked why the character believes it

6 Qualitative Insights from Responses

Especially when it comes to answering the acid test of the field, the false belief test, sometimes, in their reasoning, models generate phrases like "this is a classic example of a false belief test". Such meta-knowledge could constitute a heuristic to solve tasks by pattern matching rather than by actual solid ToM capabilities, violating the 'mentalizing' principle proposed by [10]. We do not believe this fully explains the observed performance, as this heuristic alone would not account for the improvement in other types of ToM tasks; a proper discussion on this topic can be found in the Discussion section of the paper.

Below additional qualitative analyses are provided to illustrate typical reasoning patterns by LLMs.

6.1 Filtering the facts

Often, in order to better understand a particular story, models produce sequences of 'factual' observations about the evolving story narrative, making it easier to filter for the required information. The following excerpt from a reasoning summary from Claude illustrates this pattern:

Let me analyze what happened in the story:

1. *Hannah arrives late to lunch and finds a free chair next to Ama and Matthias*
2. *Hannah asks if she can sit there*
3. *Ama says she was saving the seat for Abi*
4. *Hannah then sits with Carolyn instead*
5. *Hannah tells Carolyn that Abi went home sick that morning*
6. *Carolyn reveals that Ama was in a rehearsal all morning with her, so Ama probably didn't know Abi had gone home sick*

Now, at the end, Ama claims:

- *She's in Matthias' class*
- *She normally saves a seat for Abi*
- *But not today because Abi was sick*
- *She told all this to Hannah*

Let's check if this is correct:

- *We don't know if Ama is in Matthias' class from the story*
- *She does normally save seats for Abi (Carolyn confirms this)*
- *But according to the story, Ama DID try to save the seat for Abi (she told Hannah "I was saving that seat for Abi")*
- *Ama didn't know Abi was sick because she was in rehearsal all morning (according to Carolyn)*
- *Ama did NOT tell Hannah that she wasn't saving the seat because Abi was sick - she actually told Hannah she WAS saving the seat for Abi*

So Ama's statement is incorrect. She actually DID try to save the seat for Abi (not knowing Abi was sick), and she did NOT tell Hannah that she wasn't saving it because Abi was sick.

6.2 Perspective Taking

Perspective Taking is described in [?] as a prompting strategy to elicit ToM capabilities in LLMs, reasoning models seem to be able to, as an emergent skill, to use a perspective-taking step when needed without being prompted to do so explicitly. Let's take as an example Claude's reasoning summary when solving an instance of an imposing stories problem:

Let me carefully analyze what the story tells us about what different people believe:

1. *Hannah believes that Ama thinks Matthias has a crush on Hannah (not that Hannah likes Matthias) - Hannah says: "She thinks her boyfriend Matthias has a crush on me, but it isn't true!"*
2. *The question asks: Did Ama think that Matthias believed that Hannah liked him? So we need to determine what Ama thought about what Matthias believed about Hannah's feelings.*

From the story:

- *Hannah thinks Ama is jealous*
- *Hannah thinks Ama believes Matthias has a crush on Hannah (meaning Matthias likes Hannah)*
- *But the question is asking about whether Ama thought Matthias believed Hannah liked him (the reverse direction)*

The story doesn't actually tell us what Ama truly thinks. We only know what Hannah believes about Ama's thoughts. And specifically, Hannah believes that Ama thinks Matthias likes Hannah, NOT that Ama thinks Matthias believes Hannah likes him.

There's no information in the story that indicates Ama thought Matthias believed Hannah liked him. The only belief mentioned is Hannah's belief that Ama thinks Matthias has a crush on Hannah.

So the answer should be: No

6.3 Non-thinking Claude gets confused, reconsidering helps

Let's analyze an example:

Prompt: *There is a shoe box, Jake puts kids' toys inside of it. After that he closes the box and writes a label in it. He can't see what is inside the bag. He then looks at the box and reads 'nike shoes'. He believes that the box contains ...*

Answer: *Jake believes that the box contains Nike shoes.*

Even though Jake himself put kids' toys in the box, once he closes it and can no longer see inside, he reads the label that says "nike shoes." The label would override his memory of what he put inside, leading him to believe what the label states - that the box contains Nike shoes.

This is a classic example of how external information (the written label) can influence belief, even when it contradicts one's own recent actions.

Reprompt: *Why?*

Answer 2: *You're right to question this! Let me reconsider.*

Jake would most likely believe the box contains kids' toys because:

1. ***He just put them there himself** - This is a very recent, direct memory of his own action*
2. ***He wrote the label himself** - He knows the label is incorrect because he's the one who wrote "nike shoes" after putting toys inside*
3. ***Personal experience trumps labels** - When you personally do something moments ago, you don't forget it just because of a misleading label you yourself created*

The only way Jake would believe it contains Nike shoes is if: - Significant time had passed and he forgot - He had some memory impairment - Someone else had written the label (but the scenario says HE writes it) So Jake would believe the box contains kids' toys, despite the "nike shoes" label, because he has direct, recent knowledge of putting them there and writing a false label.

It's possible to see that, although he originally responds wrongly, asking the reasoning behind his answers (even though he already spontaneously provided a justification in the first case), leads the model to reconsider his positions and arrive at the correct conclusion. In situations like this, a point was still awarded to the model. It is hypothesized that this is not due to the model not having the ability to solve the problems, but rather by a lack of robustness in finding the correct solution path, as further discussed in the discussion section of the paper.

6.4 Model is undecided.

Several times, reasoning traces show the models considering many different explanations for a possible phenomenon. In this case, the reasoning trace of grok-3-mini when being prompted about the reason behind a sarcastic comment of a character is used as an example. Only the fundamental part is included, as Grok-3-mini includes the total of its reasoning tokens, which is too large.

(...)

Possible reasons:

1. ***He hasn't noticed the flour mess yet.** He might have entered and seen the cleaned parts but not the flour on the floor.*
2. ***Sarcasm.** He could be joking because of the mess.*
3. ***Misobservation.** He might think it's clean based on initial glance.*
4. ***Positive reinforcement.** He wants to encourage Kim despite the accident.*

(...)

6.5 Model fails to properly separate the question from the story.

This happens with some frequency in the imposing memories and in Deepseek R1's responses in general. Here is an example of the R1's reasoning, most of it is filtered, as it keeps going back and forth:

First, the scenario: There's a shoe box with kids' toys inside, but you can't see inside. The label says "nike shoes." Jake's friend told him that there are kids' toys inside and to ignore the label. Jake believed his friend. Then Jake finds the box, reads the label, and gets delighted because he's a shoe enthusiast.

The question is whether it's true or false that Jake is delighted.

Jake believed his friend, so he knows that there are kids' toys inside, not shoes. But when he sees the label, which says "nike shoes," he might forget or disregard what his friend said because he's a shoe enthusiast. However, the problem says that Jake believed his friend, so he should still believe that there are toys inside.

But the key point is that Jake is delighted when he reads the label. Since he believes his friend, he knows that the label is misleading, so he shouldn't be delighted by the label because he knows it's not true. But he is a shoe

enthusiast, so the thought of shoes might excite him even if he knows it's not real.

(...)

However, the question is about his emotional reaction. It says he is delighted to find it. But if he knows it's not shoes, why would he be delighted? Unless he is delighted by the box itself or the possibility, but that seems unlikely.

(...)

So, based on belief, he knows there are toys inside, so he shouldn't be delighted about shoes.

Therefore, it might be false that he is delighted.

(...)

But the narrative explicitly says he is delighted, so from the story, it is true that he is delighted.

(...)

So, since he reads the label and is delighted, it is consistent that he is delighted because he is not ignoring the label.

Therefore, it is true that he is delighted.

(...)

Yes, that makes sense.

Therefore, Jake is delighted to find it, so True.

6.6 Model gets it completely wrong

The only cases where models get results completely wrong are in the modifications 2A and 2B. Here is an example of the reasoning summary of Claude in problem 2B.1:

Let me trace through this step by step:

- 1. Initial state: Adu, Lin, flower, suitcase, and chest are in the room*
- 2. Adu takes the flower and puts it on the chest*
- 3. Adu leaves for school*
- 4. While Adu is away, Lin moves the flower from the chest to the suitcase*
- 5. Lin leaves for swimming class*
- 6. Adu returns from school*

The key point here is that Adu doesn't know what happened while he was away. From Adu's perspective, the last thing he knew was that he put the flower on the chest. He has no knowledge that Lin moved it.

This is a classic false belief task - Adu will have a false belief about where the flower is because he doesn't know it was moved. He will look for it where he left it.

Therefore, Adu will look for the flower on the chest.

One could see this and think that the model absorbed spurious correlations from the training data, which makes it respond to it incorrectly, as it even says

that this is a 'classic false belief test'. We argue, however, that the difference from the "classic false belief test" in the two cases where models can't solve the tests (including this one) requires a mental visualization of the task for the difference to be trivial.

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